Development of an Explainable Prolog-Based Expert System for Industrial Chemical Reactor Fault Diagnosis and Corrective Maintenance Planning

MSc KRR Mini-Project 19

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Abstract

[TO COMPLETE: 200–300 words written *after* results exist, covering items (i)–(xi) of the brief. Key facts already fixed: jacketed CSTR, 12 fault classes, 18 Horn rules plus abductive and consistency-based reasoning in strictly Prolog, 286 synthetic cases (87 dev / 199 test), explanation facility and maintenance planner.]

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1 Introduction

1.1 Engineering context

A jacketed continuous stirred-tank reactor (CSTR) couples kinetics, heat removal, mixing, pressure and feed quality. A loss of heat removal raises temperature, which raises the Arrhenius rate constant and heat generation, so a thermal excursion can become a runaway [3, 1]. Other faults degrade conversion and selectivity, deactivate catalyst or threaten containment. [TO COMPLETE: 8–12 sources for context.]

1.2 Fault-diagnosis and KRR context

Detection flags deviation, diagnosis maps symptoms to causes, prognosis predicts progression, and corrective maintenance chooses an action. A knowledge-based system is appropriate where rules, engineering constraints and auditable explanations matter [7]. Diagnosis is treated as deduction ($KB \cup E \models F$), abduction ($D \cup KB \models Obs$) and consistency-based reasoning [5, 6] over Horn clauses executed by SLD resolution [8, 9, 10]. Unlike a statistical classifier, every conclusion has a proof.

1.3 Problem statement, gap, aim

Heterogeneous, noisy, incomplete and conflicting observations must be mapped to interacting faults with an auditable chain to a maintenance action. [TO COMPLETE: research-gap table (threshold alarms, model-based, ML, expert systems, KRR) from the literature review.] **Aim.** Build and test an explainable Prolog expert system that diagnoses reactor faults and recommends corrective maintenance. [TO COMPLETE: answer the 12 research questions and 22 objectives in Section 7.]

1.4 Scope and limitations

Synthetic steady-state data only; see Section 7. Synthetic values are never presented as plant measurements.

2 Theoretical Foundation and Knowledge Representation

2.1 Reactor, assumptions and operating envelope

Liquid-phase, first-order exothermic reaction A→products in a perfectly mixed jacketed CSTR at steady state; $V_r = 2\,\mathrm{m}^3$, $\dot{m}_f \in \{0.7, 1.0, 1.3\}\,\mathrm{kg/s}$ (low/nominal/high), $T_r \approx 350\,\mathrm{K}$ held by jacket flow, $\Delta H_r = -150\,\mathrm{kJ/mol}$, $E_a = 60\,\mathrm{kJ/mol}$, $C_{A,in} = 2000\,\mathrm{mol/m}^3$. Under faults the controller output is held (manual/saturated). All parameters are engineering assumptions.

2.2 Fault taxonomy

$\mathcal{F} = \{F_K, F_T, F_H, F_C, F_M, F_Q, F_P, F_J, F_S, F_I, F_R, F_N\}$. F_U is merged into F_C/F_J because it has no independent signature with this sensor set.

Id	Meaning	Causes	Symptoms
F_K	Catalyst degradation	poisoning, sintering	$X\downarrow$, $T\downarrow$, duty$\downarrow$
F_T	Feed-temperature excursion	pre-heater fault	$T_{in}\uparrow$, $T\uparrow$, $X\uparrow$

Id	Meaning	Causes	Symptoms
F_H	Heat-transfer loss	jacket fouling	$T \uparrow$, $X \uparrow$; supply, feed, flow normal
F_C	Cooling fault	chiller/utility	$T_{j,in} \uparrow$, $T_j \uparrow$, $T \uparrow$
F_M	Mixing fault	drive, impeller	$N_a \downarrow$, $I_m \downarrow$, vibration$\uparrow$
F_Q	Feed quality	blending, contamination	$C_{A,in} \downarrow$, pH$\downarrow$, $X \downarrow$
F_P	Abnormal pressure	blocked vent, inerts	P_r above $P_{exp}(T)$
F_J	Jacket leak/circulation	pump, leak	$P_j \downarrow$, $\dot{m}_j \downarrow$, $T_j \uparrow$
F_S	Seal fault	seal wear	level$\downarrow$, $r_P \downarrow$, vibration$\uparrow$
F_I	Temperature sensor	drift, bias	$T_r \neq T_{out}$
F_R	Feed-flow control	valve, transmitter	$\dot{m}_f \neq$ load reference
F_N	Normal	–	all in band

2.3 Observation vector and variable dictionary

Measured: $T_r, T_{in}, T_{out}, T_{j,in}, T_j, P_r, \dot{m}_f, C_{A,in}, C_A, C_B, C_P, Q_r, V_r, N_a, P_j, \dot{m}_j$, pH, V_{RMS}, I_m. Derived by the evidence layer: e_T, e_P, X, S, residuals r_P, r_T, ρ_E. c_p is specific heat, C_P product concentration; no symbol C_p is used (the brief's Eq. 20 clash). Full dictionary: `docs/KB_SPECIFICATION.md`. [TO COMPLETE: paste as table (Appendix C).]

2.4 Mathematical formulation

Material balance and kinetics.

$$\tau = V_r / \dot{V}_f \tag{1}$$

$$\dot{V}_f = \dot{m}_f / \rho \tag{2}$$

$$\frac{dC_A}{dt} = \frac{C_{A,in} - C_A}{\tau} - r_A \tag{3}$$

$$r_A = a C_A^n, \; n = 1 \text{ (rate } kC_A\text{)} \tag{4}$$

$$k = a\, k_0 \exp\big(-E_a/(RT_r)\big) \tag{5}$$

$$X = \frac{C_{A,in} - C_A}{C_{A,in}} \tag{6}$$

$$X = \frac{k\tau}{1 + k\tau} \text{ (steady, first order)} \tag{7}$$

$$S = \frac{C_P}{C_{A,in} - C_A} \tag{8}$$

$$C_A + C_B + C_P = C_{A,in} \text{ (closure)} \tag{9}$$

Energy balance and heat transfer.

$$\rho V_r c_p \frac{dT_r}{dt} = \dot{m}_f c_p (T_{in} - T_r) + (-\Delta H_r) r_A V_r - \dot{Q}_j \tag{10}$$

$$\dot{Q}_{gen} = \dot{Q}_r + \dot{Q}_j \text{ (steady)} \tag{11}$$

$$\dot{Q}_{gen} = (-\Delta H_r) \frac{\dot{m}_f}{\rho} C_{A,in} X \tag{12}$$

$$\dot{Q}_r = \dot{m}_f c_p (T_{out} - T_{in}) \tag{13}$$

$$\dot{Q}_j = \varepsilon \, \dot{m}_j c_{pj} (T_r - T_{j,in}) \tag{14}$$

$$\varepsilon = 1 - \exp\big(-UA/(\dot{m}_j c_{pj})\big) \tag{15}$$

$$\dot{Q}_j = \dot{m}_j c_{pj} (T_j - T_{j,in}) \text{ (measured)} \tag{16}$$

$$\Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)} \tag{17}$$

$$UA = \left[\frac{1}{h_i A_i} + R_f + \frac{1}{h_o A_o} \right]^{-1} \tag{18}$$

$$Re = \frac{\rho v D_h}{\mu} \tag{19}$$

$$Pr = \frac{c_p \mu}{k_f} \tag{20}$$

$$Nu = \frac{h D_h}{k_f} \tag{21}$$

$$\rho_E = \frac{\dot{Q}_{gen} - \dot{Q}_r - \dot{Q}_j}{\dot{Q}_{gen}} \tag{22}$$

$$\Delta T_{ad} = \frac{(-\Delta H_r) C_{A,in}}{\rho c_p} \tag{23}$$

$$\frac{d\dot{Q}_{gen}}{dT} > \frac{d\dot{Q}_{rem}}{dT} \text{ (runaway condition)} \tag{24}$$

Mechanical, pressure and leak relations.

$$P_{agit} = N_p \rho N^3 D^5 \tag{25}$$

$$\omega = 2\pi N_a / 60 \tag{26}$$

$$\dot{W}_m = \tau_m \omega \tag{27}$$

$$I_m = I_0 + I_1 (N_a / N_{a,0})^2 \tag{28}$$

$$P_r V_g = n_r R T_r \tag{29}$$

$$P_{exp} = P_{atm} + P_v(T) \approx 101.3 + 8e^{0.045(T-300)} \tag{30}$$

$$r_P = P_r - P_{exp}(T_{best}) \tag{31}$$

$$\frac{dV_r}{dt} = \dot{V}_{in} - \dot{V}_{out} - \dot{V}_{leak} \tag{32}$$

$$\dot{m}_{j,out} = \dot{m}_{j,in} - \dot{m}_{leak} \tag{33}$$

$$r_T = |T_r - T_{out}| \tag{34}$$

Deviation, states and evidence aggregation.

$$d_x = \frac{x - x_{ref}}{x_{ref}} \tag{35}$$

$$\epsilon_T = \frac{T_r - T_{ref}}{T_{ref}}, \quad \epsilon_P = \frac{P_r - P_{ref}}{P_{ref}} \tag{36}$$

$$z_x = \frac{x - \mu_x}{\sigma_x} \tag{37}$$

$$\text{state}(x) = \begin{cases} \text{low} & d_x < \ell_x \\ \text{high} & d_x > u_x \\ \text{normal} & \text{otherwise} \end{cases} \tag{38}$$

$$E(F_k) = \sum_i w_i e_i(F_k) \tag{39}$$

$$C(F_k) = E(F_k) / \textstyle\sum_i w_i \tag{40}$$

$$P(F_k|E) \propto P(E|F_k)P(F_k) \tag{41}$$

$$\text{Score}(D) = \sum_{v \in Ex(D)} w_v + \lambda|R(D)| - \mu|D| \tag{42}$$

Logic and diagnosis.

$$KB \cup E \models F \tag{43}$$

$$D \cup KB \models Obs \tag{44}$$

$$KB \cup Obs \cup D \not\models \bot \tag{45}$$

$$D^* = \arg\min_D |D| \tag{46}$$

$$KB \cup Obs \cup D \models Maint \tag{47}$$

$$D \text{ hits every conflict set } CS_j \text{ (Reiter)} \tag{48}$$

Evaluation.

$$\text{Acc} = \tfrac{TP+TN}{TP+TN+FP+FN} \tag{49}$$

$$\text{Prec} = \tfrac{TP}{TP+FP} \tag{50}$$

$$\text{Rec} = \tfrac{TP}{TP+FN} \tag{51}$$

$$F_1 = \tfrac{2\,\text{Prec}\,\text{Rec}}{\text{Prec}+\text{Rec}} \tag{52}$$

$$\text{Cov} = N_{definite}/N_{test} \tag{53}$$

$$\text{FAR} = N_{normal \to fault}/N_{normal} \tag{54}$$

$$\text{ExplRate} = N_{explained}/N_{diagnosed} \tag{55}$$

$$\text{PlanValidity} = N_{valid\,plans}/N_{test} \tag{56}$$

The set above contains 56 numbered equations (target: ≥ 40); the weights w rank hypotheses only and never decide membership.

2.5 KRR representation

Layers: `state/3` (evidence) → `cond/2` (engineering conditions) → `r/3` (diagnostic rules). Natural language → FOL → Horn → Prolog, e.g. "a quenched reaction with normal feed quality indicates catalyst degradation":

$$\forall c: \; LowX(c) \wedge LowT(c) \wedge \neg LowFeedConc(c) \wedge \neg LowpH(c) \to Fault(c, f_k)$$

```
cond(C, activity_loss_pattern) :- cond(C, low_conversion), cond(C, low_reactor_temperature).
r(rk01, fk, C) :- cond(C, activity_loss_pattern), ( observed(C, cain) ; observed(C, ph) ),
                  \+ cond(C, low_feed_concentration), \+ cond(C, low_feed_ph).
```

Negation as failure expresses exceptions, but only over *observed* evidence, so missing data never silently counts as "normal". Defaults: normal operation unless a primary variable is abnormal; nominal feed assumed when $C_{A,in}$ is unmeasured.

2.6 Three reasoning modes

Deduction: `fired/3` resolves `r/3` goals by SLD. **Abduction:** subset-minimal sets of ≤ 3 faults whose `explains/3` effects cover all coverable primary symptoms. **Consistency:** a hypothesis is rejected when an observed variable contradicts every necessary effect (`must_any/2`) or violates `exclusive/2`. [TO COMPLETE: one worked formal derivation and one Prolog trace for each mode: generate with `demos` in `run_all`.]

3 Design and Development of the Prolog Expert System

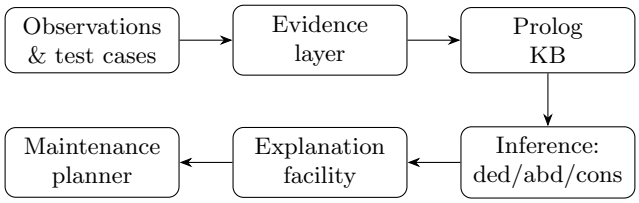


Figure 1: Layered architecture (modules: params, evidence, kb, reasoning, explain, planner, eval).

3.1 Rule inventory, knowledge acquisition

18 rules (rk01, rq01–02, rh01, rc01–02, rt01, rj01–02, rm01–03, rs01–02, rp01, rr01–02, ri01); provenance and basis in `docs/KB_SPECIFICATION.md` §3. [TO COMPLETE: extend with the specific papers each rule relies on (literature → knowledge → rule → predicate → test).]

3.2 Explanation facility and maintenance planner

`explain/1` reports observed evidence, derived conditions, rules fired, diagnosis with evidence confidence, competing/rejected hypotheses and a maintenance plan (fault, evidence, action, urgency, safety constraint, verification). `sld_trace/2` prints the proof tree. [TO COMPLETE: insert one explanation trace from `results.txt`.]

4 Dataset, Experimental Methodology and Evaluation

4.1 Dataset

Synthetic only: steady-state model with ε-NTU jacket (Eqs. 7, 14, 15), Gaussian sensor noise, fixed seed 20260101, 286 cases: A normal 24, B single 132, C pairs 30, D ambiguous 24, E incomplete 30, F noisy/borderline 31, G sensor+physical 15. Dev/test split 87/199; tolerances tuned on dev only. **Limitation L1:** generator and KB share the same physics, so results measure internal consistency, not plant performance.

4.2 Results

[TO COMPLETE: paste tables from `results.txt`: reasoning-mode comparison, per-category results, per-class table, confusion matrix, false-alarm rate, mean/max time, explainability rate, plan validity; ablation (rule groups, evidence classes, tolerance scale, KB growth).]

5 Software Documentation and Reproducibility

Environment SWI-Prolog 9.x; commands and layout in `README.md`; traceability matrix in `docs/KB_SPECIFICATI` §6. [TO COMPLETE: record exact SWI version, OS, and final test output.]

6 Assessment Framework and Deliverables

[TO COMPLETE: required tables/figures checklist; contributions; future work (real/plant data, dynamics, more sensors, UI).]

7 Conclusion

[TO COMPLETE: synthesise against research questions after results; must not restate the introduction.]

Limitations to state. Synthetic and circular data; steady state only; parsimony can mask a weaker fault; sensor-fault detection limited to the T_r/T_{out} pair; $\leq$ 3 simultaneous faults; thresholds are assumptions; maintenance text requires process-safety review.

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[TO COMPLETE: expand to 70–90 unique sources managed in EndNote, APA 7, $\geq 60\%$ recent peer-reviewed. This list of 13 is a verified starting set only.]